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Class: 4C

1. Import libraries

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
```

```
plt.style.use('ggplot')
```

2. Load dataset

```
df = pd.read_csv("https://drive.google.com/uc?id=1klwIj4oQT43DWBVH5KWz0Z5BhcyGz-cR")
df.head()
```

	Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quantity	Tax 5%	Sales	Date	Time	Payment	cogs
0	750-67-8428	Alex	Yangon	Member	Female	Health and beauty	74.69	7	26.1415	548.9715	1/5/2019	1:08:00 PM	Ewallet	522.83
1	226-31-3081	Giza	Naypyitaw	Normal	Female	Electronic accessories	15.28	5	3.8200	80.2200	3/8/2019	10:29:00 AM	Cash	76.40
2	631-41-3108	Alex	Yangon	Normal	Female	Home and lifestyle	46.33	7	16.2155	340.5255	3/3/2019	1:23:00 PM	Credit card	324.31
3	123-19-1176	Alex	Yangon	Member	Female	Health and beauty	58.22	8	23.2880	489.0480	1/27/2019	8:33:00 PM	Ewallet	465.76
4	373-73-7910	Alex	Yangon	Member	Female	Sports and travel	86.31	7	30.2085	634.3785	2/8/2019	10:37:00 AM	Ewallet	604.17

Next steps: Generate code with df New interactive sheet

```
cols_to_drop = ["Invoice ID", "Date", "Time", "gross income", "Tax 5%", "cogs", "gross margin percentage"]
df = df.drop(cols_to_drop, axis=1)
```

3. EDA (Correlation + Understanding)

```
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 10 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Branch         1000 non-null   object
 1   City           1000 non-null   object
 2   Customer type  1000 non-null   object
 3   Gender         1000 non-null   object
 4   Product line   1000 non-null   object
 5   Unit price     1000 non-null   float64
 6   Quantity       1000 non-null   int64
 7   Sales          1000 non-null   float64
 8   Payment        1000 non-null   object
 9   Rating         1000 non-null   float64
dtypes: float64(3), int64(1), object(6)
memory usage: 78.3+ KB
```

	Unit price	Quantity	Sales	Rating
count	1000.000000	1000.000000	1000.000000	1000.00000
mean	55.672130	5.510000	322.966749	6.97270
std	26.494628	2.923431	245.885335	1.71858
min	10.080000	1.000000	10.678500	4.00000
25%	32.875000	3.000000	124.422375	5.50000
50%	55.230000	5.000000	253.848000	7.00000
75%	77.935000	8.000000	471.350250	8.50000
max	99.960000	10.000000	1042.650000	10.00000

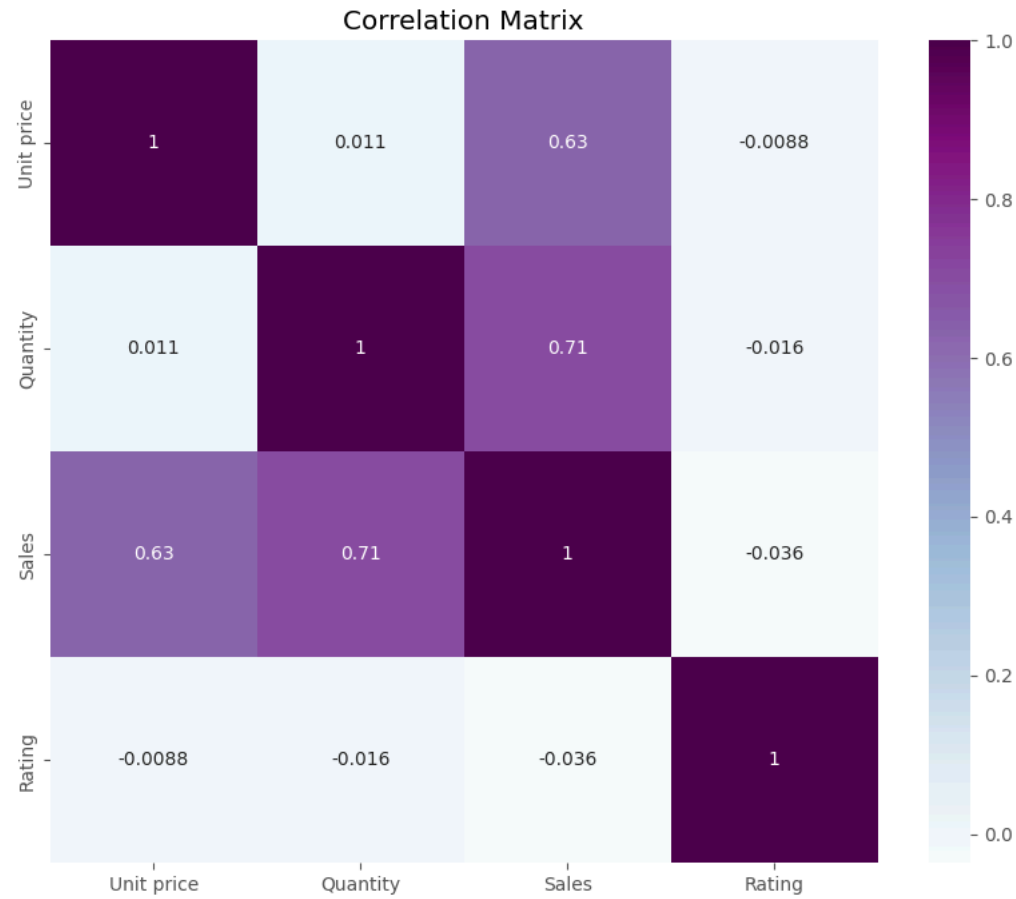
```
df.isnull().sum()
```

	0
Branch	0
City	0
Customer type	0
Gender	0
Product line	0
Unit price	0
Quantity	0
Sales	0
Payment	0
Rating	0

dtype: int64

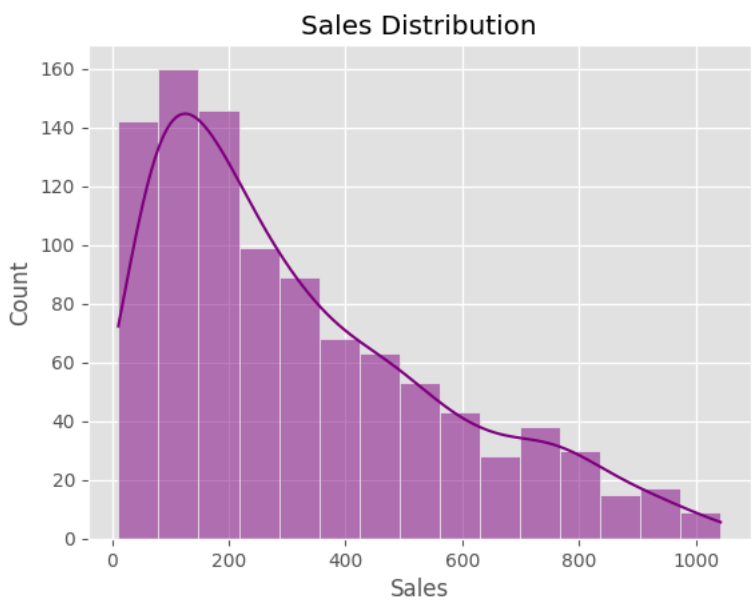
Correlation heatmap

```
plt.figure(figsize=(10,8))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="BuPu")
plt.title("Correlation Matrix")
plt.show()
```



Salesm distribution

```
sns.histplot(df["Sales"], kde=True, color="purple")
plt.title("Sales Distribution")
plt.show()
```



4. Feature Engineering

```
# Convert categorical text into dummy variables
df = pd.get_dummies(df, drop_first=True)
```

5. Split features and target

```
X = df.drop('Sales', axis=1)
y = df['Sales']
```

6. Train/Test split (80/20)

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
print("Train shape:", X_train.shape)
print("Test shape:", X_test.shape)
```

```
Train shape: (800, 16)
Test shape: (200, 16)
```

7. Scaling

```
scaler = StandardScaler()
# Fit only on training data to prevent leakage
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

8. Train model + Predict

```
# Train model
model = LinearRegression()
model.fit(X_train_scaled, y_train)

y_pred = model.predict(X_test_scaled)

# Calculate Metrics
r2 = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))

# Cross-Validation to ensure the model is robust
cv_r2 = cross_val_score(model, X_train_scaled, y_train, cv=5).mean()

print(f"R2 Score: {r2:.4f}")
print(f"Cross-Validated R2: {cv_r2:.4f}")
print(f"MAE: {mae:.2f}")
print(f"RMSE: {rmse:.2f}")
```

```
R2 Score: 0.9019
Cross-Validated R2: 0.8817
MAE: 59.38
RMSE: 79.90
```

9. Model Evaluation

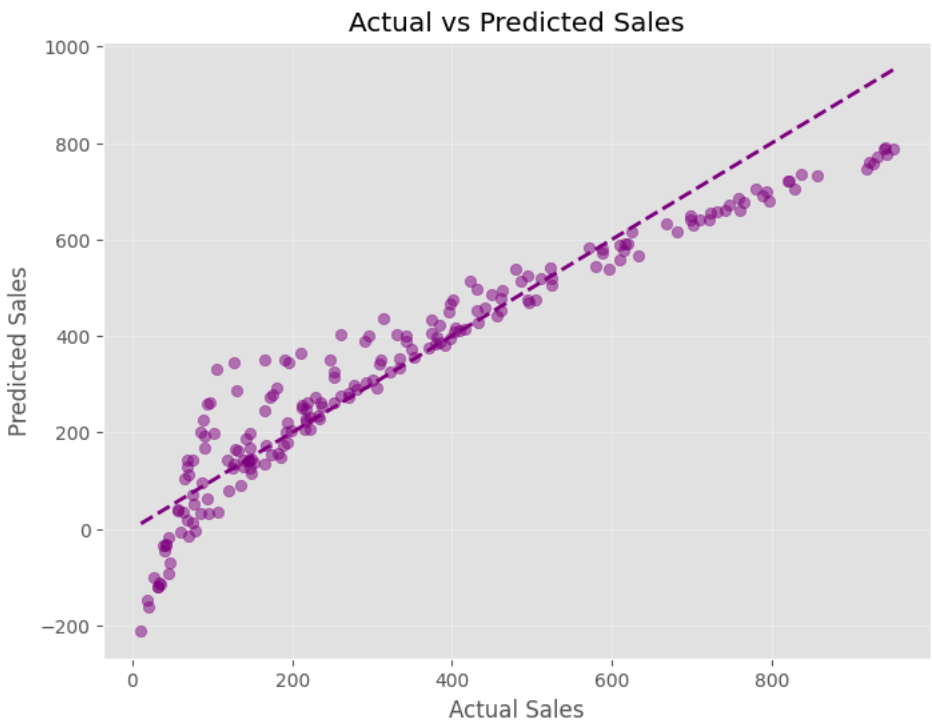
```
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
mae = mean_absolute_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("Model Performance Metrics:")
print("R2 Score:", r2)
```

```
print("MAE:", mae)
print("RMSE:", rmse)
```

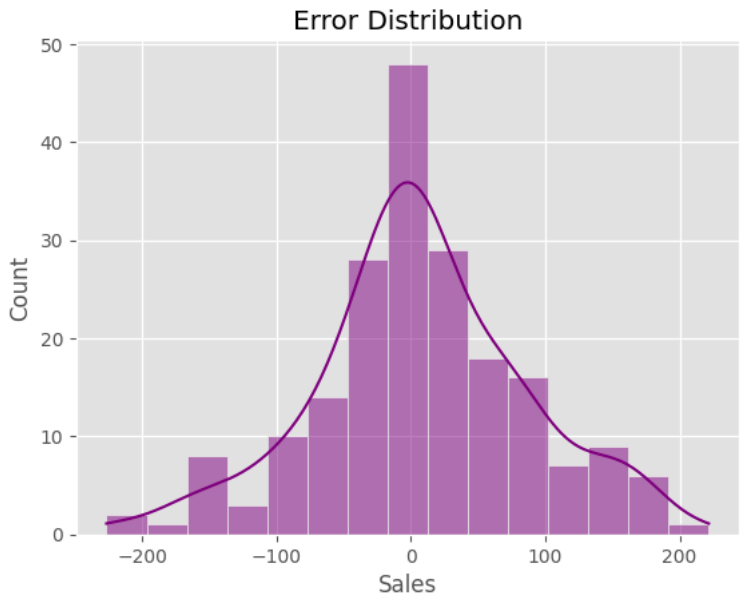
```
Model Performance Metrics:
R2 Score: 0.9018802054495287
MAE: 59.38214157910874
RMSE: 79.89728615964347
```

10. Display Results

```
plt.figure(figsize=(8,6))

plt.scatter(y_test, y_pred, alpha=0.5, color="purple")

plt.plot(
    [y_test.min(), y_test.max()],
    [y_test.min(), y_test.max()],
    linestyle="--",
    color="purple",
    linewidth=2
)

plt.xlabel("Actual Sales")
plt.ylabel("Predicted Sales")
plt.title("Actual vs Predicted Sales")

plt.grid(True, alpha=0.3)
plt.show()
```




```
# Residuals
residuals = y_test - y_pred

sns.histplot(residuals, kde=True, color='purple')
plt.title("Error Distribution")
plt.show()
```


Error Distribution
Count
Sales
50
40
30
20
10
0
−200
−100
0
100
200